**This assignment will not be graded and is only for practice.**

## Level 1

1) Rank of a matrix $A$ is equal to **1 point**

- [ ] the row rank of the matrix $A$.
- [ ] the column rank of the matrix $A$.
- [ ] the nullity of the matrix $A$.
- [ ] the number of dependent variables in the system of equations $Ax = 0$.
- [ ] the number of independent variables in the system of equations $Ax = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

the row rank of the matrix $A$.
the column rank of the matrix $A$.
the number of dependent variables in the system of equations $Ax = 0$.

2) Let nullity of the matrix $A_{3\times5}$ be 2. Find the rank of the matrix $A_{3\times5}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

3) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

4) If $A$ is an $m \times n$ matrix, then which of the following statements is true? ***1 point***

- ○ $rank(A) + nullity(A) = m$
- ○ $rank(A) + nullity(A) = n$
- ○ $rank(A) + nullity(A) = mn$
- ○ $rank(A) + nullity(A) = max\{m, n\}$
- ○ $rank(A) + nullity(A) = min\{m, n\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$rank(A) + nullity(A) = n$

5) Which of the following is true for a homogeneous system of linear equations $Ax = 0$? ***1 point***

- ☐ It may have no solution.
- ☐ It can have infinitely many solutions.
- ☐ It can have a unique solution.
- ☐ Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

No, the answer is incorrect.

Score: 0

Accepted Answers:

It can have infinitely many solutions.
It can have a unique solution.
Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

6) Which of the following options are correct for a square matrix $A$ of order $n \times n$, where $n$ is any natural number? ***1 point***

- [ ] If the determinant is non-zero, then the nullity of $A$ must be 0.
- [ ] If the determinant is non-zero, then the nullity of $A$ may be non-zero.
- [ ] If the nullity of $A$ is non-zero, then the determinant of $A$ must be 0.
- [ ] If the nullity of $A$ is non-zero, then the determinant of $A$ may be non-zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the determinant is non-zero, then the nullity of $A$ must be 0.
If the nullity of $A$ is non-zero, then the determinant of $A$ must be 0.

7) Choose the set of correct statements. ***1 point***

- [ ] If nullity of a $3 \times 3$ matrix is $c$ for some natural number $c$, $0 \leq c \leq 3$, then the nullity of $-A$ will also be $c$.
- [ ] $nullity(A + B) = nullity(A) + nullity(B)$
- [ ] Nullity of the zero matrix of order $n \times n$, is $n$.
- [ ] Nullity of the zero matrix of order $n \times n$, is $0$.
- [ ] There exist square matrices $A$ and $B$ of order $n \times n$, such that nullity of both $A$ and $B$ is $0$, but the nullity of $A + B$ is $n$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If nullity of a $3 \times 3$ matrix is $c$ for some natural number $c$, $0 \leq c \leq 3$, then the nullity of $-A$ will also be $c$.
Nullity of the zero matrix of order $n \times n$, is $n$.
There exist square matrices $A$ and $B$ of order $n \times n$, such that nullity of both $A$ and $B$ is $0$, but

the nullity of $A + B$ is $n$.

8) The rank of an $n \times n$ matrix all of whose entries are equal to 1 is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

9) If $A$ is a $3 \times 4$ matrix, then which of the following options are true? ***1 point***

- [ ] rank(A) must be less than or equal to 3.
- [ ] nullity(A) must be greater than or equal to 1.
- [ ] If $A$ has $2$ columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these $2$ columns, then $nullity(A) = 2$.
- [ ] If $A$ has $2$ columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these $2$ columns, then $nullity(A) = 1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

rank(A) must be less than or equal to 3.
nullity(A) must be greater than or equal to 1.
If $A$ has $2$ columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these $2$ columns, then $nullity(A) = 2$.

**Level 2**

10) Let $Ax = 0$ be a homogeneous system of linear equations which has infinitely many solutions, where $A$ is an $m \times n$ matrix (where, $m > 1, n > 1$). Which of the following statements are possible? ***1 point***

- [ ] $rank(A) = m$ and $m < n$.
- [ ] $rank(A) = m$ and $m > n$.
- [ ] $rank(A) = m$ and $m = n$.
- [ ] $nullity(A) = n$.
- [ ] $nullity(A) = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$rank(A) = m$ and $m < n$.
$nullity(A) = n$.
$nullity(A) = 0$.

Consider the coefficient matrix $A$ of the following system of linear equations to answer questions 11 and 12:

$$3x_1 + 2x_2 + x_3 = 0$$

$$x_1 + x_3 = 0$$

11) Which one of the following vector spaces represents the null space of $A$ appropriately? ***1 point***

- $\{(-t, t, t) \mid t \in \mathrm{R}\}$
- $\{(t_1, t_2, \frac{t_2 - t_1}{2}) \mid t_1, t_2 \in \mathrm{R}\}$
- $\{(t, -t, t) \mid t \in \mathrm{R}\}$
- $\{(t_1, t_2, \frac{t_1 + t_2}{2}) \mid t_1, t_2 \in \mathrm{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-t, t, t) \mid t \in \mathrm{R}\}$

12) What will be the rank of $A$ and nullity of $A$? ***1 point***

- $rank(A) = 3, nullity(A) = 2$
- $rank(A) = 2, nullity(A) = 1$
- $rank(A) = 1, nullity(A) = 2$
- $rank(A) = 2, nullity(A) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$rank(A) = 2, nullity(A) = 1$

13) Let $Ax = 0$ be a homogeneous system of linear equations which has a unique solution, where $A \in \mathrm{R}^{n \times n}$. What is the nullity of $A$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

***1 point***

14) Suppose $x_1 = 0$ solves $Ax = 0$, where $A \in \mathrm{R}^{2 \times 4}$. What is the minimum number of elements in a linearly independent subset of null space of $A$ that also spans the set of solutions of $Ax = 0$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

15) Consider the following sets: ***1 point***

$S_1 = \{(1, 2, 3), (4, 5, 6), (8, 9, 10)\}$
$S_2 = \{(1, 2, 1), (2, 4, 2), (1, 2, 5)\}$
$S_3 = \{(1, 2, 3), (4, 5, 6), (7, 8, 10)\}$
$S_4 = \{(1, 2), (4, 5)\}$
$S_5 = \{(1, 2), (4, 5), (3, 8)\}$

Choose the set of correct options

- [ ] $S_1$ is a basis of $\mathrm{R}^3$.
- [ ] $S_2$ is a basis of $\mathrm{R}^3$.
- [ ] $S_3$ is a basis of $\mathrm{R}^3$.
- [ ] $S_4$ is a basis of $\mathrm{R}^2$.
- [ ] $S_5$ is a basis of $\mathrm{R}^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S_3$ is a basis of $\mathrm{R}^3$.
$S_4$ is a basis of $\mathrm{R}^2$.

Your score is: 0/15